

A Decision-Theoretic Approach to Monetary Policy under Weak Identification

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Introduction

- ▶ DSGE models with different structural parameters observationally equivalent
 - ▶ empirics discipline parameters with moments
 - ▶ here focus on impulse responses (Christiano et al., 2005)
- ▶ The parameter-to-moment mapping may be many-to-one
 - ▶ Chari et al. (2009): reduced-form fit can mask different structural models
 - ▶ Mavroeidis et al. (2014); Canova and Sala (2009): limited information can leave some parameters weakly identified
- ▶ Common practice: evaluate policy at point estimate (Schmitt-Grohé and Uribe, 2007; Galí, 2015)
 - ▶ disregards parameter uncertainty
 - ▶ can be misleading when ψ is weakly identified

This Paper

Question: how should a policymaker choose one rule when welfare-relevant parameters are (potentially) weakly identified?

- ▶ Treat policy choice as a statistical decision problem
 - ▶ action: operational interest-rate rule
 - ▶ state: structural parameter
 - ▶ loss: welfare cost of the rule
- ▶ Apply "optimal decision rules for weak gmm" (Andrews and Mikusheva, 2022)
 - ▶ use a Bayes rule
 - ▶ posterior makes trade-off between structural values explicit
 - ▶ choose the rule with lowest posterior expected loss
- ▶ Show policy consequences in two settings
 - ▶ Monte Carlo: regret relative to the oracle policy rule
 - ▶ medium-scale NK model: rule disagreement on US data

Relation to Robust Policy

- ▶ Hansen–Sargent robustness (Hansen and Sargent, 2008)
 - ▶ problem: misspecification around a benchmark model
 - ▶ response: choose against least-favorable distortions
- ▶ Giannoni–Woodford robustness (Giannoni and Woodford, 2005, 2017)
 - ▶ problem: optimal policy depends on uncertain model dynamics
 - ▶ response: target criteria that remain valid across a class of model dynamics
- ▶ This paper
 - ▶ problem: weak identification of welfare-relevant parameters
 - ▶ response: choose a rule by posterior expected welfare

Structure of Presentation

Policy Problem and Weak Identification

Decision-Theoretic Response

Empirical Evidence

Conclusion

The Policy Problem

- ▶ If ψ were known, the policymaker would solve

$$a^*(\psi) \in \arg \max_{a \in \mathcal{A}} W(a, \psi).$$

- ▶ a : interest-rate policy
 - ▶ ψ : policy-invariant structural parameters
 - ▶ $W(a, \psi)$: welfare criterion
- ▶ The plug-in policy replaces ψ by an estimate

$$\hat{a}^{\text{PI}} = a^*(\hat{\psi}).$$

- ▶ **Problem:** plug-in evaluates policies at one point
 - ▶ disregards sampling uncertainty
 - ▶ severe when ψ is weakly identified

Impulse-Response Matching

- ▶ Empirical DSGE work often estimates θ by matching impulse responses
 - ▶ Christiano et al. (2005) discipline a medium-scale DSGE model with IRFs
 - ▶ Hall et al. (2012) methodology formalization
- ▶ Minimum-distance estimator: $\hat{\theta}^{MD} = \arg \min_{\theta \in \Theta} Q_T(\theta),$

$$Q_T(\theta) = \{\hat{\gamma}_T - g(\theta)\}' \hat{V}^{-1} \{\hat{\gamma}_T - g(\theta)\}$$

- ▶ $\hat{\gamma}_T$: empirical impulse responses
 - ▶ $g(\theta)$: model-implied impulse responses
 - ▶ $\hat{V}$: weighting matrix
- ▶ Strong identification
 - ▶ the population criterion is uniquely minimized at θ_0
- ▶ Weak identification
 - ▶ local to identification failure
 - ▶ identified set: Θ_0

Policy as a Statistical Decision Problem

- ▶ State: $\theta = (\psi, \alpha)$
 - ▶ ψ : policy-invariant structural parameters
 - ▶ α : sample policy environment
- ▶ Data: empirical impulse responses $\hat{\gamma}_T$
- ▶ Action: interest-rate policy $a \in \mathcal{A}$
- ▶ Loss: $L(a, \psi) = -W(a, \psi)$
- ▶ Decision rule: $\delta : \mathcal{X} \rightarrow \mathcal{A}$ maps data to policy
- ▶ Criterion: choose a rule with low risk (expected loss)

$$\mathbb{E}_{\theta_0} [L\{\delta(X_T), \psi\}]$$

Optimality Criterion

- ▶ Need an optimality criterion to choose a decision rule
 - ▶ no uniformly best rule under weak identification
- ▶ Admissibility
 - ▶ rules out dominated decision rules
 - ▶ many admissible rules can remain
- ▶ Minimax and minimax regret
 - ▶ hedge against the worst-case state
 - ▶ can be overly conservative
 - ▶ not guaranteed to select unique rule
- ▶ Bayes risk
 - ▶ averages risk across plausible structural values
 - ▶ implies admissibility
 - ▶ makes the trade-off across states explicit
 - ▶ under strict curvature, it delivers a unique rule

Quasi-Bayes for IRF Matching

- ▶ The Bayes policy minimizes posterior expected loss

$$\hat{a}^{\text{QB}} = \arg \min_{a \in \mathcal{A}} \int L(a, \psi) \pi(\theta | X_T) d\theta.$$

- ▶ For IRF matching, use the quasi-posterior

$$\pi(\theta | X_T) \propto \exp \left[-\frac{T}{2} Q_T(\theta) \right] \pi(\theta).$$

- ▶ Based on the MD criterion (Chernozhukov and Hong, 2003; Andrews and Mikusheva, 2022)
- ▶ Not a full posterior: no likelihood for the whole DSGE model
- ▶ Quasi-posterior uses only the IRF-matching information
- ▶ Strong identification: concentrates near $\hat{\theta}^{MD}$
- ▶ Weak identification: remains diffuse in weak directions

Monte Carlo Design

- ▶ Weak-identification design adapted from Canova and Sala (2009)

In deviations from steady state, with i.i.d. Gaussian shocks,

$$\text{IS: } y_t = \psi_1 \mathbb{E}_t y_{t+1} + \psi_2 (i_t - \mathbb{E}_t \pi_{t+1}) + e_{1t},$$

$$\text{NKPC: } \pi_t = \psi_3 \mathbb{E}_t \pi_{t+1} + \psi_4 y_t + e_{2t},$$

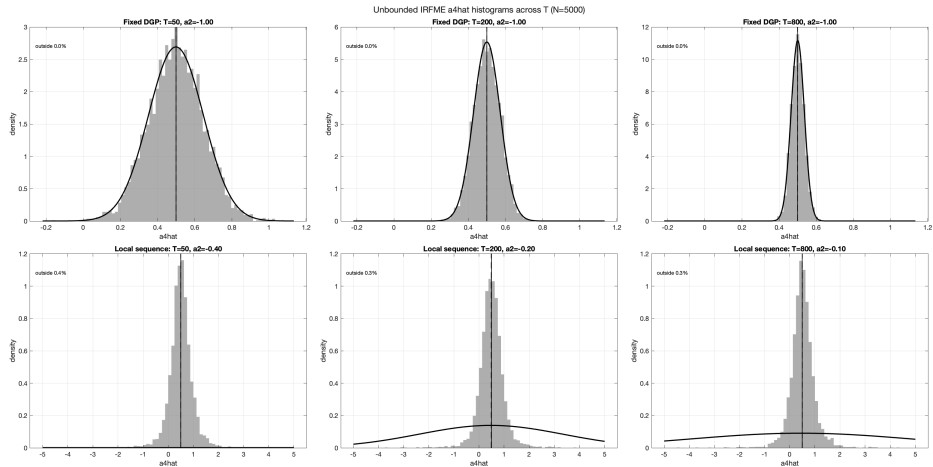
$$\text{Policy: } i_t = \psi_5 \mathbb{E}_t \pi_{t+1} + e_{3t},$$

- ▶ Dual-mandate welfare loss (Rudebusch and Svensson, 1999)

$$W(a, \psi) = -\{\text{Var}_\psi(\pi; a) + \text{Var}_\psi(y; a) + 0.1 \text{Var}_\psi(\Delta i; a)\}$$

- ▶ Welfare-relevant parameters: ψ_2 and ψ_4
- ▶ Identification regimes: strong and weak
- ▶ Benchmark: regret relative to oracle policy

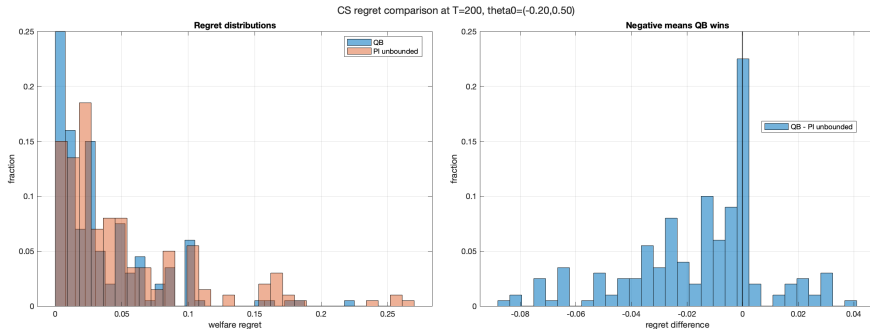
Monte Carlo: Weak Identification



Sampling distributions of $\hat{\psi}_4$ under strong and weak identification

Monte Carlo Result

- ▶ Quasi-Bayes reduces average plug-in regret by 33.4%
- ▶ Quasi-Bayes leads to higher welfare in 67.5% of samples
- ▶ The two policies differ in approximately 81% of samples



Regret relative to the oracle policy

Empirical Application: Medium-Scale NK Model

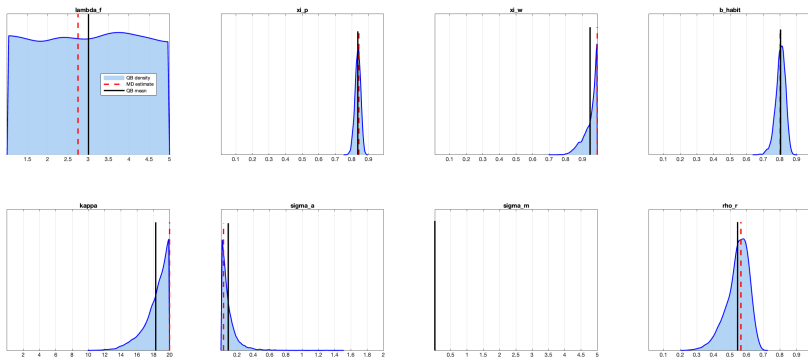
- ▶ Medium-scale New Keynesian model with standard nominal and real frictions (Christiano et al., 2005)
- ▶ Estimated by impulse-response matching to a monetary VAR
- ▶ Sample: US quarterly data, 1965:Q3–1995:Q3
- ▶ Relative IRF-matching information criterion of Hall et al. (2012) selects $H = 7$
- ▶ Policy: backward-looking Taylor policy

$$i_t = \rho_r i_{t-1} + (1 - \rho_r) \{ \varphi_\pi \pi_{t-1} + \varphi_y y_{t-1} \}, \quad a = (\varphi_\pi, \varphi_y, \rho_r)$$

- ▶ Welfare: dual-mandate

$$W(a, \psi) = -\{ \text{Var}(\pi) + \text{Var}(y) + 0.1 \text{Var}(\Delta i) \}.$$

Medium-Scale NK Model: Quasi-Posterior

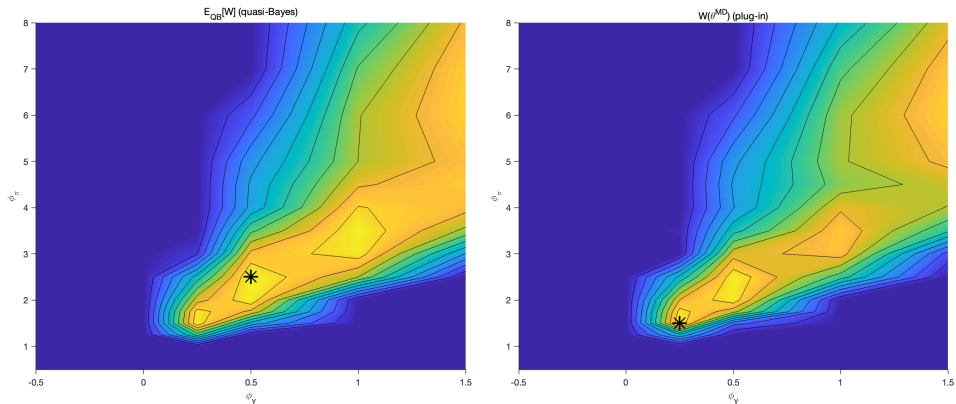


Shaded: quasi-posterior density; dashed: MD estimate; solid: QB mean

Medium-Scale NK Model: Quasi-Posterior

- ▶ Several parameters are well pinned down relative to the prior
 - ▶ price stickiness ξ_p
 - ▶ habit b
 - ▶ sample-policy smoothing ρ_r
- ▶ Other dimensions remain weakly disciplined or sit at economically meaningful bounds
 - ▶ price markup λ_f remains very diffuse
 - ▶ wage stickiness ξ_w hits its upper bound
 - ▶ investment adjustment cost κ hits its upper bound
 - ▶ monetary-shock scale σ_m hits its lower bound
- ▶ Their uncertainty is transmitted to policy through the welfare ranking over Taylor rules
 - ▶ price stickiness ξ_p and wage stickiness ξ_w change inflation and wage dynamics
 - ▶ monetary-shock scale σ_m changes the size of policy disturbances
 - ▶ price markup λ_f mainly shifts the markup block and has a limited policy effect here

Medium-Scale NK Model: Welfare Surface



Left: QB posterior-mean welfare; right: plug-in welfare at $\hat{\theta}^{MD}$

Medium-Scale NK Model: Rule Disagreement

The selected policies $a = (\varphi_\pi, \varphi_y, \rho_r)$ are

$$\hat{a}^{\text{PI}} = (1.50, 0.25, 0.90)$$

$$\hat{a}^{\text{QB}} = (2.50, 0.50, 0.95)$$

- ▶ Plug-in reflects Taylor-style inflation response with high smoothing (Taylor, 1993; Clarida et al., 2000)
- ▶ Quasi-Bayes favors stronger stabilization
 - ▶ some draws imply more persistent inflation and output dynamics
 - ▶ stronger responses reduce inflation and output volatility

▶ Backup: caveats

Conclusion

- ▶ Weak identification is a policy problem
 - ▶ weakly identified parameters can rank policies differently
- ▶ Plug-in policy evaluates welfare at one parameter value
 - ▶ sampling uncertainty is discarded before the policy step
- ▶ Quasi-Bayes carries structural uncertainty into welfare
 - ▶ choose the policy with lowest quasi-posterior expected loss
- ▶ Evidence
 - ▶ Monte Carlo: quasi-Bayes reduces regret
 - ▶ medium-scale NK model: selected policies disagree

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Backup: Weak-GMM Limit Experiment

The normalized moment process satisfies

$$m_T(\cdot) \rightsquigarrow Z(\cdot), \quad Z \sim P_\mu,$$

where Z is a Gaussian process with unknown mean function μ and consistently estimable covariance kernel.

The Gaussian limit induces a quasi-likelihood of the form

$$\mathcal{L}_Q(\theta; Z) \propto \exp\{-Q(\theta)/2\}.$$

Its sample analogue is

$$Q_T(\theta) = m_T(\theta)' \hat{\Sigma}(\theta)^{-1} m_T(\theta),$$

which yields the quasi-posterior used in the main text.

Andrews and Mikusheva (2022) show that this quasi-Bayes construction is the diffuse-prior Bayes rule in the weak-GMM limit experiment.

Backup: Caveats

- ▶ Medium-scale NK result is qualitative
 - ▶ empirical object is rule disagreement
 - ▶ not a large welfare-gain claim
- ▶ Weak directions can inherit prior weight
 - ▶ diffuse prior avoids adding tight curvature
 - ▶ prior sensitivity remains relevant under weak identification
- ▶ Quasi-Bayes credible sets are Bayesian objects
 - ▶ they need not have frequentist coverage